

```
from google.colab import files
uploaded = files.upload()
```

No file chosen Upload widget is only available when the cell has been executed in the current browser session. Please rerun this cell to enable.
 Saving gender_submission.csv to gender_submission (1).csv
 Saving test.csv to test.csv

```
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns
```

```
df = sns.load_dataset('titanic')
df.head()
```

	survived	pclass	sex	age	sibsp	parch	fare	embarked	class	who	adult_male	deck	embark_town	alive	alone
0	0	3	male	22.0	1	0	7.2500	S	Third	man	True	NaN	Southampton	no	False
1	1	1	female	38.0	1	0	71.2833	C	First	woman	False	C	Cherbourg	yes	False
2	1	3	female	26.0	0	0	7.9250	S	Third	woman	False	NaN	Southampton	yes	True
3	1	1	female	35.0	1	0	53.1000	S	First	woman	False	C	Southampton	yes	False
4	0	3	male	35.0	0	0	8.0500	S	Third	man	True	NaN	Southampton	no	True

```
df.info()
df.describe()
df.isnull().sum()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 891 entries, 0 to 890
Data columns (total 15 columns):
#   Column      Non-Null Count  Dtype
---  -
0   survived    891 non-null    int64
1   pclass      891 non-null    int64
2   sex         891 non-null    object
3   age         714 non-null    float64
4   sibsp       891 non-null    int64
5   parch       891 non-null    int64
6   fare        891 non-null    float64
7   embarked    889 non-null    object
8   class       891 non-null    category
9   who         891 non-null    object
10  adult_male  891 non-null    bool
11  deck        203 non-null    category
12  embark_town 889 non-null    object
13  alive       891 non-null    object
14  alone       891 non-null    bool
dtypes: bool(2), category(2), float64(2), int64(4), object(5)
memory usage: 80.7+ KB
```

	0
survived	0
pclass	0
sex	0
age	177
sibsp	0
parch	0
fare	0
embarked	2
class	0
who	0
adult_male	0
deck	688
embark_town	2
alive	0
alone	0

dtype: int64

```
# Fill missing values
df['age'].fillna(df['age'].median(), inplace=True)
df['embarked'].fillna(df['embarked'].mode()[0], inplace=True)
```

```
# Drop column with too many nulls
df.drop('deck', axis=1, inplace=True)
```

/tmp/ipykernel_2794/2401979740.py:2: FutureWarning: A value is trying to be set on a copy of a DataFrame or Series through c...
The behavior will change in pandas 3.0. This inplace method will never work because the intermediate object on which we are

For example, when doing 'df[col].method(value, inplace=True)', try using 'df.method({col: value}, inplace=True)' or df[col]

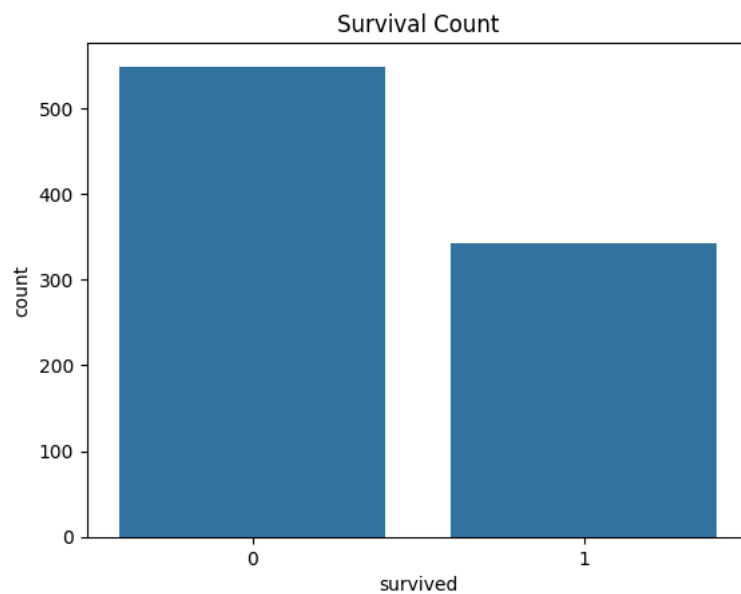
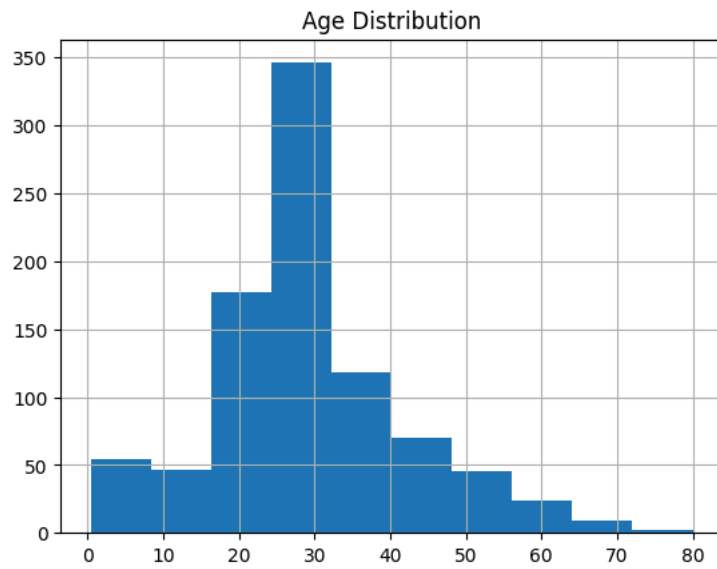
```
df['age'].fillna(df['age'].median(), inplace=True)
/tmp/ipykernel_2794/2401979740.py:3: FutureWarning: A value is trying to be set on a copy of a DataFrame or Series through c...
The behavior will change in pandas 3.0. This inplace method will never work because the intermediate object on which we are
```

For example, when doing 'df[col].method(value, inplace=True)', try using 'df.method({col: value}, inplace=True)' or df[col]

```
df['embarked'].fillna(df['embarked'].mode()[0], inplace=True)
```

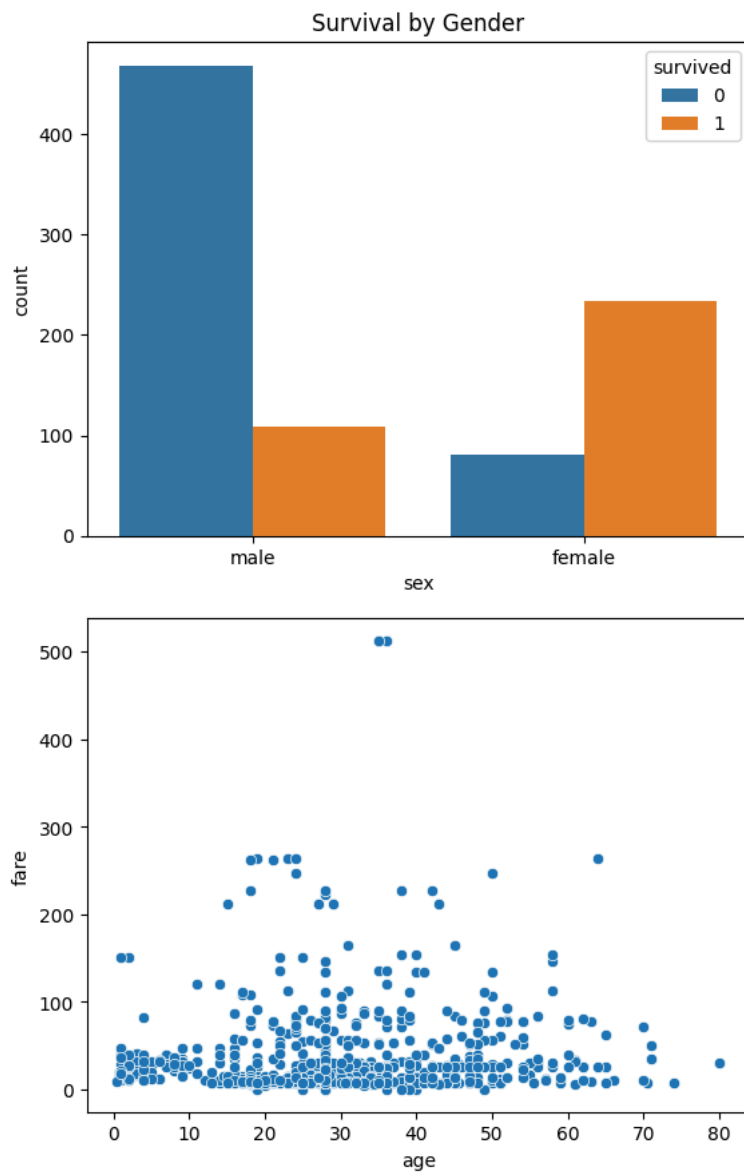
```
# Histogram
df['age'].hist()
plt.title("Age Distribution")
plt.show()
```

```
# Countplot
sns.countplot(x='survived', data=df)
plt.title("Survival Count")
plt.show()
```

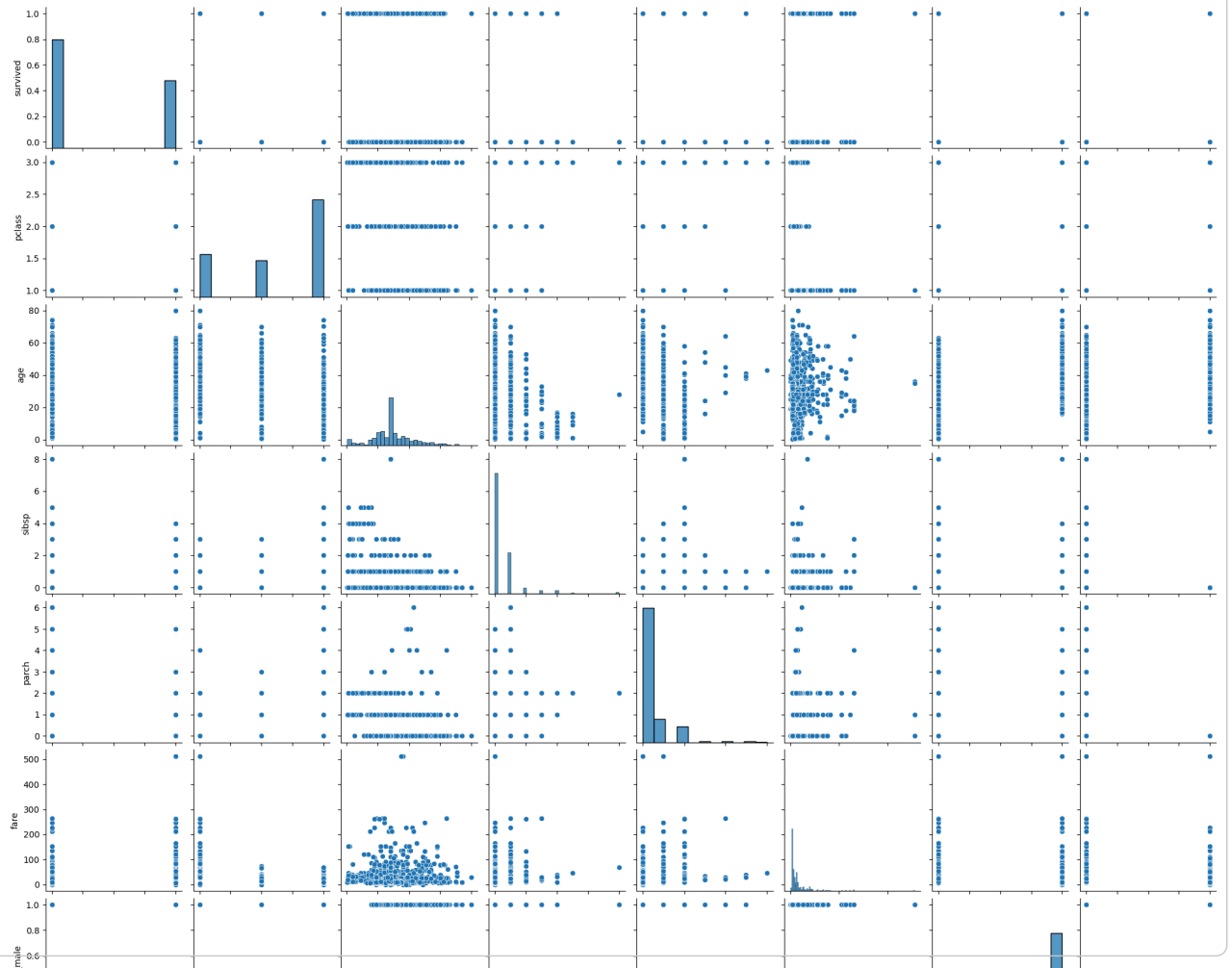


```
# Survival vs Gender
sns.countplot(x='sex', hue='survived', data=df)
plt.title("Survival by Gender")
plt.show()

# Scatterplot
sns.scatterplot(x='age', y='fare', data=df)
plt.show()
```



```
sns.pairplot(df)
plt.show()
```



```
plt.figure(figsize=(10,6))
sns.heatmap(df.corr(numeric_only=True), annot=True, cmap='coolwarm')
plt.title("Correlation Heatmap")
plt.show()
```

